

Vincent Lam

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Computer Science graduate building full-stack and security-focused software with automated tests, CI and live demos. Background in data validation and secure document workflows for the State of California.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, R

Frameworks & data: FastAPI, Next.js, React, Node.js, SQLAlchemy, Pydantic, MySQL, SQLite, IndexedDB

Security: Access control (ABAC, separation of duties), OIDC / JWT, Web Crypto (AES-256-GCM, PBKDF2), HMAC audit logs, CSP

AI & tools: Claude API (tool use, structured outputs), Git, GitHub Actions, Docker, pytest, Playwright, Selenium, Linux

PROJECTS

Aegis – Just-in-time access control engine | Python, FastAPI, SQLAlchemy, Claude API

[Live demo](#)

- Built an IAM service where employees request temporary access in plain English: Claude parses each request into a schema-validated policy, and an attribute-based rules engine allows or denies it with a reason per rule.
- Added separation-of-duties approvals, time-boxed break-glass access, automatic revocation on expiry or role change, and an HMAC-SHA256 hash-chained audit log that detects tampering.
- Identity only from verified JWTs (OIDC/JWKS); 60 pytest tests in CI; the live demo runs the real API in-browser via WebAssembly.

Privacy Blocker – Local-first data-broker opt-out tool | JavaScript, Web Crypto, Python

[Live demo](#)

- Encrypted all personal data in the browser (PBKDF2-SHA256, 600k iterations, then AES-256-GCM) with idle auto-lock; a CSP of connect-src 'none' keeps the app from sending data anywhere.
- Tracks opt-outs for about 40 brokers and drafts CCPA requests; a scheduled GitHub Actions job emails brokers, logging no personal data.

Installous – AI stock research app | TypeScript, Next.js, Claude API

[Code](#)

- Five-factor stock scoring model and a Claude advisor that calls Zod-validated tools for live quotes, fundamentals and news.

Nagare – Japanese reader that runs entirely in the browser | TypeScript, Next.js, IndexedDB

[Live demo](#)

- Tokenizes text in the browser (kuromoji.js) against a 22k-word JMDict dictionary with JLPT levels; translation runs on-device.

W.E.M.U 2.0 – Senior project, team of eight | Node.js, Selenium IDE

- Web app for the California Department of Consumer Affairs (3.4M licenses, 36 entities) to request, approve and track form migrations across development, test and production databases.
- Handled the production side of form migrations and the contact forms page; Selenium IDE suites covered authentication, migrations and role-based access.

Movie Recommendation Service – Team project (Scrum) | Java, Jersey, Docker, MySQL

- REST API with full CRUD for movies, Academy Award queries by year and category, and query-based recommendations.

EXPERIENCE

Office of Legislative Counsel – *Office Technician (Typing)*

Sept 2026 – Present

- Produce drafts, revisions and final publications of legislation in specialized automated document-workflow systems, and deliver them to clients and the Office of State Publishing through secure channels.
- Partner with IT staff to diagnose and resolve system issues; write the manuals and SOPs used to train staff.

California Department of Health Care Services

Student Assistant – Public Hospital Redirection Payments & Veterans Homes Programs

Jan 2025 – June 2025

- Validated and reconciled claims and financial reports for state and federal reimbursement programs, resolving discrepancies with fiscal and policy teams before payment release.
- Entered and cross-checked program data with high accuracy, reducing analyst correction time and improving audit readiness.

Student Assistant – SharePoint Program & Records Support

Nov 2022 – April 2024

- Ran QA checks on department-wide SharePoint content for accuracy, metadata integrity and accessibility compliance.
- Resolved access issues, broken links and version conflicts, and standardized libraries with templates and structured filing.

California Housing Finance Agency – *Student Assistant, Loan Operations & Data Quality*

May 2022 – Oct 2022

- Refined Python and R scripts used for internal data validation, reducing manual review time; ran data-quality checks on loan documents and system entries.

EDUCATION

California State University, Sacramento – B.S. Computer Science

Summer 2025